BST Construction from Given Elements

Algorithm

Step 1: Start.

Step 2: Define a node structure containing data, a left pointer, and a right pointer.

Step 3: Initialize the root of the BST as NULL.

Step 4: Insert the elements 14, 15, 4, 9, 7, 18, 3, and 5 in the given order using the BST insertion rule: smaller values go to the left and larger values go to the right.

Step 5: After each insertion, display the current BST.

Step 6: Display the final BST using Inorder (Left, Root, Right), Preorder (Root, Left, Right), and Postorder (Left, Right, Root) traversals.

Step 7: Calculate and display the height of the BST by finding the longest path from the root to a leaf. Use the same height convention consistently (for example, count edges).

Step 8: Count and display the leaf nodes, which are nodes with no left or right child.

Step 9: Ask the user for a value to search. Traverse the BST by comparing the target with each node; report whether it is found.

Step 10: Ask the user for a value to delete. If the node is a leaf, remove it; if it has one child, replace it with that child; if it has two children, replace it with its inorder successor and remove the successor.

Step 11: Display the updated BST after deletion.

Step 12: Stop.